

UNIVERSITY DEPARTMENT OF BOTANY SIDO KANHU MURMU UNIVERSITY, DUMKA

Head of The Department

University Department of Botany
S.K.M. University, Dumka (Jharkhand)
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Letter No- SKMU/PG/Bot/79/2024

Date- 08/05/24

To,

The D.S.W.,
Sido Kanhu Murmu University, Dumka.

Through,

The NEP Co-ordinator,
SKMU Dumka.

Sub: Regarding submission of syllabus of mushroom culture as minor subject of vocational paper as per NEP.

Sir,

In response to your letter no SKMU/DSW/61/24, Dt. 04/05/2024, I hereby submit the required Syllabi for your needful.

Thanking You.

Yours Sincerely,

08/05/24

Head of the Department,
University Department of Botany,
Sido Kanhu Murmu University, Dumka.

Enclosures: As above.

HEAD
P.G. Department of Botany
S.K.M. University, Dumka

SIDO KANHU MURMU UNIVERSITY, DUMKA



Syllabus for Minor Paper from Vocational as per NEP.
For Semester-II to VIII

FOR UNDER GRADUATE COURSE

*Implemented from
Academic Session 2022-2026*

Naraj Basker

Imaya

S. S. Saha
08/09/24

Table No. 1: Marks distribution pattern (25 Marks for each credit)

Subject	Semester	Code	Credits	Full Marks	Semester Internal Examination			End Semester University Examination	
					T	A	FM	T	P
Minor Courses	IV	MN-2B	4	100	10	5	15	60	25
Minor Courses	VI	MN-2C	4	100	10	5	15	60	25
Minor Courses	VIII	MN-2D	4	100	10	5	15	60	25

Instruction to Question Setter:

For Semester Internal Examination (SIE 10 marks, 1Hr Exam): There will be two group of questions. Question No.1 will be very short answer type in Group A consisting of five questions of 1 mark each. Group B will contain descriptive type two questions of five marks each, out of which any one to be answered.

For End Semester Examination (ESE 60 marks, 3Hrs Exam): There will be two group of questions. Group A is compulsory which will contain three questions. Question No.1 will be very short answer type consisting of five questions of 1 mark each. Question No.2 & 3 will be short answer type of 5 marks.

Group B will contain descriptive type five questions of fifteen marks each, out of which any three are to answered.

Neeraj Baskar

Omaya

Prasanna
08/05/24

Syllabus for Minor Paper from Vocational as per NEP.

Sem-IV

Paper 2nd

Full Marks: 100

Internal Marks: 10+05=15

Theory Marks: 60

Course Learning Outcomes:

1. Understand the basics of mushroom
2. Identify edible and poisonous mushroom.
3. Understand nutritional and medicinal values of mushroom.

1. **Unit-1: Introduction;** General Characters, reproduction and classification of Fungi.

08 classes

2. **Unit-2.** Definition, history, types, identification, nutritional and medicinal values of mushrooms; identification of edible and poisonous mushrooms. Types of edible mushrooms available in India.

14 classes

3. **Unit-3.** Life history and cultivation and nutritional value of *Volvariella volvacea*, *Pleurotus citrinopileatus*, *Agaricus bisporus*.

14 classes

Practical: Marks:25

24 classes

1. Collection, staining and identification,
2. preparation of slides and identification of mushrooms
3. Morpho-histological study of edible and poisonous mushrooms like *Volvariella volvacea*, *Pleurotus citrinopileatus*, *Agaricus bisporus* etc.
4. Spotting
5. Viva- voce
6. Records/projects/ charts/ models etc

Suggested readings:

1. Philip G. Miles, Shu-Ting Chang, 1997. Mushroom biology, World Scientific, Singapore.
2. Kaul, T.N. 1999. Introduction to mushroom science, Oxford & IBH Co., Pvt. Ltd., New Delhi
3. Bahl, N. 1988. Handbook on mushrooms. Oxford & IBH Publishing Co., Pvt. Ltd., New Delhi.
4. Dube H.C. Fungi (4th Edition). Scientific Publishers.

Nancy Baskin

Omaya

Anisha
08/05/24

Syllabus for Minor Paper from Vocational as per NEP.

Sem-VI

Paper 3rd

Full Marks: 100

Internal Marks: 10+05=15

Theory Marks: 60

Course Learning Outcomes:

1. Sterilize apparatus, tools, media and chamber.
2. Prepare the culture media, mushroom bed and unit
3. Prepare mother spawn, daughter spawn and cultivate mushroom.
4. Develop entrepreneurship skill in cultivating mushroom.
5. Analyse cost benefit ratio.

1. **Unit-1:** Study of tools and techniques like- inoculation hook, inoculation loop, low-cost stove, sieve, culture rack, mushroom unit (Thatched house) water sprayer, tray, pure culture; medium, sterilization, preparation of spawn, multiplication. Mushroom bed preparation – paddy straw. **18 Classes**

2. **Cultivation technology;** Infrastructure, substrates (locally available) polythene bag, vessels, inoculation hook, inoculation loop, low-cost stove, sieves, culture rack, mushroom unit (thatched house) water sprayer, tray, small polythene bag, pure culture; medium, sterilization, preparation of spawn, multiplication, mushroom bed preparation- sugarcane trash. Factors affecting the mushroom bed preparation- low-cost technology, composting technology in mushroom production. **18 Classes**

Practical: Marks:25

24 classes

1. Study of tools and techniques.
2. Sterilization of glassware and tools.
3. Preparation of Culture Media
4. Sterilization of Culture media,
5. Inoculation and culture of mushroom.
6. Preparation of Spawn.
7. Preparation of Mushroom Bed.
8. Spotting
9. Viva- voce
10. Records/projects/ charts/ models etc

Suggested readings:

5. Philip G. Miles, Shu-Ting Chang, 1997. Mushroom biology, World Scientific, Singapore.
6. Kaul, T.N. 1999. Introduction to mushroom science, Oxford & IBH Co., Pvt. Ltd., New Delhi
7. Bahl, N. 1988. Handbook on mushrooms. Oxford & IBH Publishing Co., Pvt. Ltd., New Delhi.
8. Dube H.C. Fungi (4th Edition). Scientific Publishers.

Neeraj Basker

Indya

Shamha
28/05/24

Syllabus for Minor Paper from Vocational as per NEP.

Sem-VIII

Paper 4th

Full Marks: 100

Internal Marks: 10+05=15

Theory Marks: 60

Course Learning Outcomes:

1. Store the mushroom produced.
 2. Prepare food for consumption.
 3. Analyse the cost benefit ratio.
 4. Marketing of the product can be done.
 - 5.
1. **Unit-1: Storage and nutrition;** short term storage (refrigeration upto 24 hours), long term storage (canning, pickles, papads), drying, storage in salt solution. Nutrition; proteins, amino acids, mineral elements nutrition- carbohydrates, crude fibre content-vitamins.
20 Classes
2. **Unit-2: Food preparation;** types of foods prepared from mushroom. Research centres- National level and regional level. Cost benefit ratio- Marketing in India and abroad, export value
16 Classes

Practical: Marks:25

24 classes

1. Methods of short term storage (Refrigeration – up to 24 hours). Long term storage (Canning, pickles, papads), drying, storage in salt solutions analysis of proteins- amino acids, mineral elements carbohydrates, crude fibre content and vitamins.
2. Methods of preparations of foods prepared from mushroom.
3. Evaluation of Cost benefit ratio, marketing in India.
4. Spotting
5. Viva- voce
6. Records/projects/ charts/ models etc

Suggested readings:

1. Philip G. Miles, Shu-Ting Chang, 1997. Mushroom biology, World Scientific, Singapore.
2. Kaul, T.N. 1999. Introduction to mushroom science, Oxford & IBH Co., Pvt. Ltd., New Delhi
3. Bahl, N. 1988. Handbook on mushrooms, Oxford & IBH Publishing Co., Pvt. Ltd., New Delhi.
4. Dube H.C. Fungi (4th Edition). Scientific Publishers.

Neejay Baskar.

Amay

Shreemha
08/05/24

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Letter No. SKMU/DSW/...61.../24

date 04/05/2024

To

Dr. T.P. Singh
NEP Co-ordinator
HOD, Economics, SKMU, Dumka

Subject- Regarding preparing of syllabus of Minor from Vocational as per NEP.

Sir,

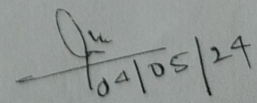
With reference to above subject, you are requested to arrange the syllabus of following subjects (Minor from Vocational) for Sem IV, VI and VIII with the help of subject concerned teachers. Since students of 2022-26 have already chosen the following subjects in semester I.

1. Entrepreneurship Development Skill
2. Awareness of Rights and Laws
3. Drawing and Colour Studies
4. Mushroom Cultivation
5. Digital Marketing and Management
6. Advertising Management
7. Fundamentals of Computers
8. Fundamentals of Accounting
9. Critical Thinking and Writing
10. Disaster Management
11. Issues of Rural Government

Semester	Paper No	Marks Distribution
Sem IV	Paper II	25 + 75
Sem VI	Paper III	25 + 75
Sem VIII	Paper IV (Dissertation)	25 + 75

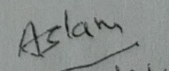
Kindly submit the syllabus at DSW Office within 7 working days.

It is for your kind information and needful action.


DSW

Copy to:

1. Controller of examination, SKMU, Dumka
2. PA to VC/Registrar
3. Guard File and University website


04/05/24